Machine Learning Model Deployment Assignment

OVERVIEW OF ASSIGNMENT

COURSE : CAPSTONE PROJECT

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Objective

- Deploy a machine learning model
- Ensure scalability and accessibility
- Use REST API for predictions

Project Overview

- Select ML problem and dataset
- Deploy on AWS, GCP, or Azure
- Focus on **deployment and operationalization**

Assignment Steps

Phase 1: Problem & Data

- Select a dataset and a corresponding machine learning problem that interests you. (datasets on platforms like Kaggle, UCI, data.gov).
- Perform exploratory data analysis.
- Understand data characteristics.

Phase 1: Preprocessing

- Handle missing values.
- Feature scaling and encoding.
- Save preprocessing pipeline.

Phase 1: Model Training

- Train multiple models.

  (at least two different types of ML models)

- Evaluate performance metrics.

  (e.g., accuracy, precision, recall, F1-score, MAE, R^2).

- Select best model.

Phase 2: Cloud Setup

- Create cloud account
- Set up project/resources
- Use free tier services

Phase 2: Containerization

- Create Dockerfile
- Use FastAPI or Flask
- Test container locally

Phase 2: Cloud Deployment

- Deploy your containerized application to a cloud service. Some recommended services include:
  - **AWS**: Elastic Beanstalk, Amazon SageMaker, or Elastic Kubernetes Service (EKS) for a more advanced option.
  - **GCP**: Cloud Run, Google AI Platform, or Google Kubernetes Engine (GKE).
  - **Azure**: Azure App Service, Azure Machine Learning, or Azure Kubernetes Service (AKS).
- Configure the deployment to be scalable and resilient.

Phase 2: API Endpoint Configuration

- Expose a secure API endpoint that can receive POST requests with input data in JSON format and return model predictions.
- Implement basic API key authentication to secure your endpoint.

Final Report & Submission:

➢ 1-Project report with screenshots

- An **Executive Summary** of your project.
- A description of the **Machine Learning Problem** and the dataset used.
- An overview of your **Model Development Process (**preprocessing, training, and evaluation.)
- A detailed **Deployment Architecture Diagram** & an explanation of the cloud services you used.
- A section on **Challenges Faced** and how you overcame them.
- A **Conclusion** summarizing your work and suggesting future improvements.
- **Note: your report MUST include screenshots of your working deployment for full marks.**

Final Report & Submission:

2. Code Repository:

- A link to a private GitHub or GitLab repository containing:
- All **Jupyter notebooks** or Python scripts used for data analysis and model training.
- The final trained **model artifact** and preprocessing pipeline.
- Your **application code** (e.g., main.py for FastAPI/Flask).
- The Dockerfile and any other configuration files (e.g., requirements.txt).
- A comprehensive README.md file.

Final Report & Submission:

3. Live API Endpoint:

A URL to your deployed and functional API endpoint, along with any necessary API keys for testing.

Good Videos

- 56:27 How To: Use AWS SageMaker for End-to-End ML — Setup, Training, Deployment, and Bias Detection
- 44:23 From Laptop to Cloud: A Complete End-to-End Data Science Project with AWS SageMaker
- 22:44 End-to-End Machine Learning Project Using AWS SageMaker
- 40:25 End To End Machine Learning Project Implementation Using AWS Sagemaker
- 3:04:11 End-to-End MLOps Pipeline with AWS SageMaker, GitHub Actions, MLflow & FastAPI | Resume Project 2026